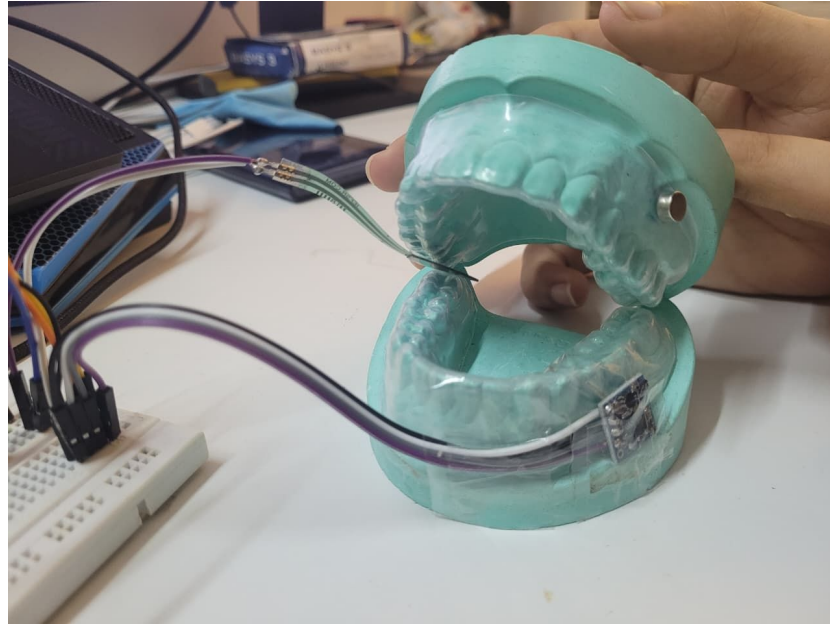




An Intra-Oral Wearable for Sleep Bruxism Monitoring

Magnetometer-Based Jaw Tracking + Per-Tooth Force Sensing

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Clinical Motivation

Why We Need Better Bruxism Monitoring

Sleep bruxism affects millions but remains underdiagnosed. Dentists currently rely on subjective reports, visual wear patterns, and nightguard inspection—methods that lack precision and quantification.

Existing solutions like polysomnography and EMG chin straps are expensive, obstructive, or impractical for routine clinical use.

What's Missing

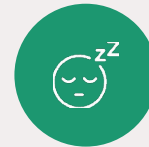
- Objective jaw movement intensity measurement
- Quantified grinding event frequency
- Side-specific tooth pressure data
- Affordable, patient-friendly monitoring

Stakeholders & Impact



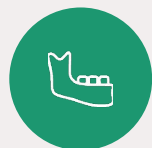
Dentists

Primary users who gain objective data for treatment planning, splint design, and patient counseling



Sleep Specialists

Enhanced diagnostic capability as polysomnography add-on for comprehensive sleep disorder assessment



TMJ Specialists

Orthodontists and pain specialists treating temporomandibular disorders and alignment issues



Patients

Comfortable home monitoring with actionable insights into nighttime grinding behavior

Target Use-Case

Our system provides **overnight sleep bruxism monitoring** using an Invisalign-style intra-oral device that automatically detects and quantifies jaw activity throughout the night.



Grinding Detection

Lateral jaw movement patterns characteristic of bruxism events



Chewing-Like Motion

Rhythmic jaw activity during sleep stages



Baseline Position

Rest state calibration for movement comparison



Pressure Distribution

Per-tooth force mapping via FSR array



System Architecture

Dual-Sensor Hybrid Design

- 1

Jaw Motion Tracking

NdFeB-coated magnet mounted on upper teeth tracks 3D movement via magnetometer + ESP32 on lower appliance
- 2

Per-Tooth Force Sensing

FSR sensors embedded under appliance capture pressure distribution; expandable to multi-tooth PCB array
- 3

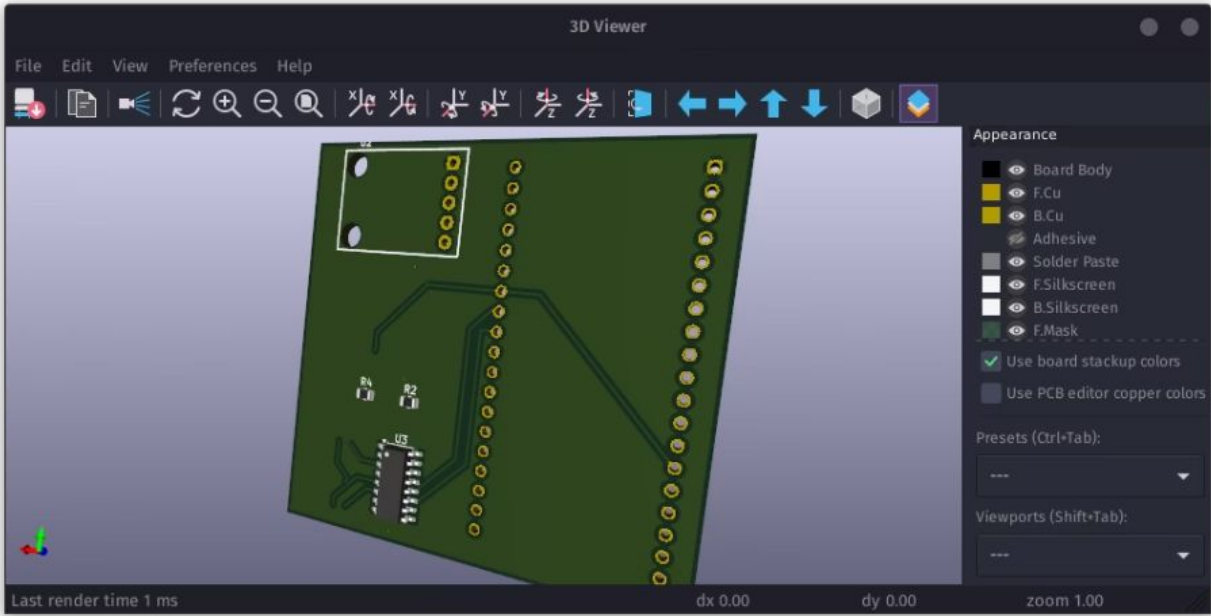
Wireless Telemetry

BLE streaming to smartphone or laptop with offline data storage and analysis
- 4

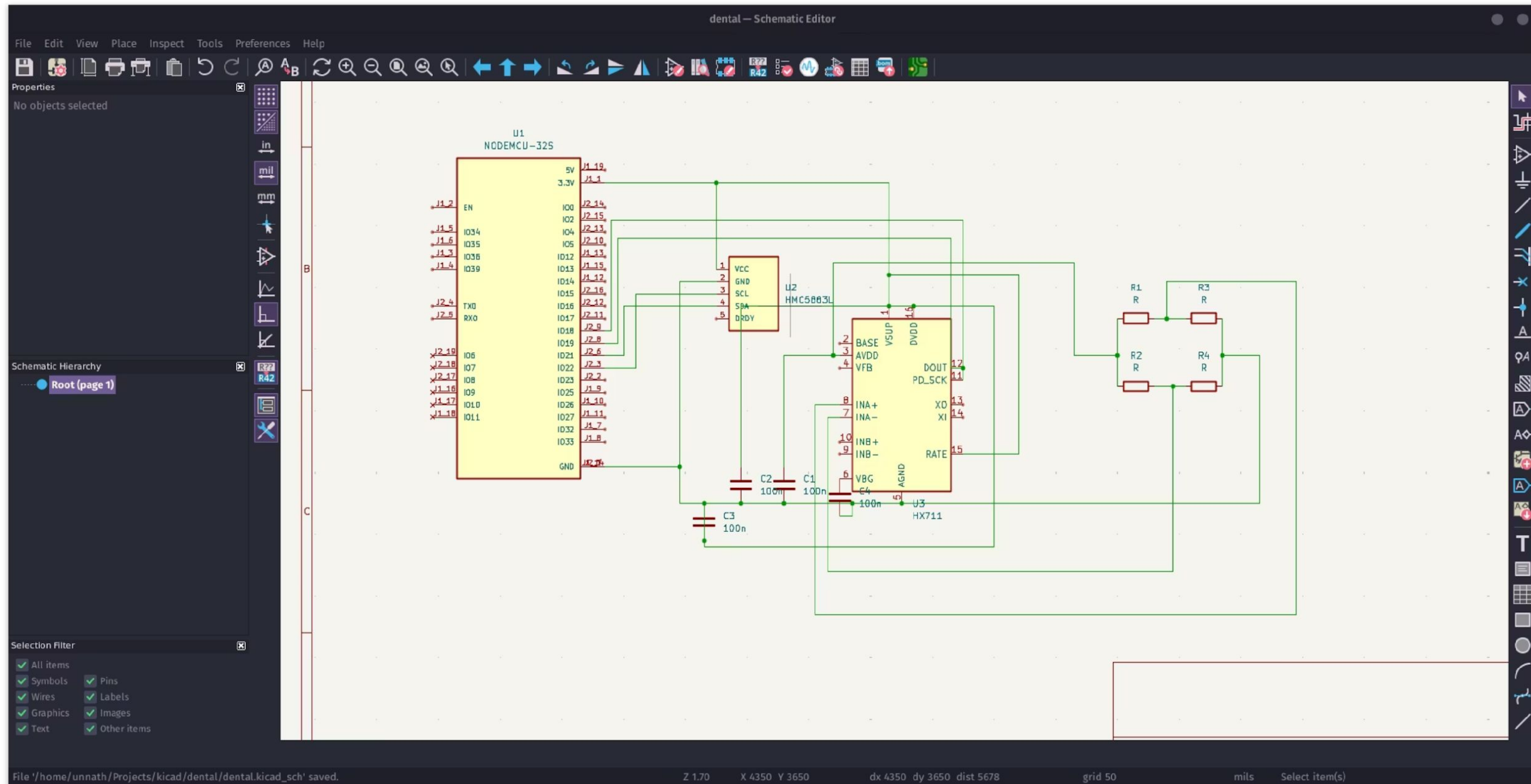
Power Management

Battery optimized for full-night monitoring exceeding 8 hours of continuous operation

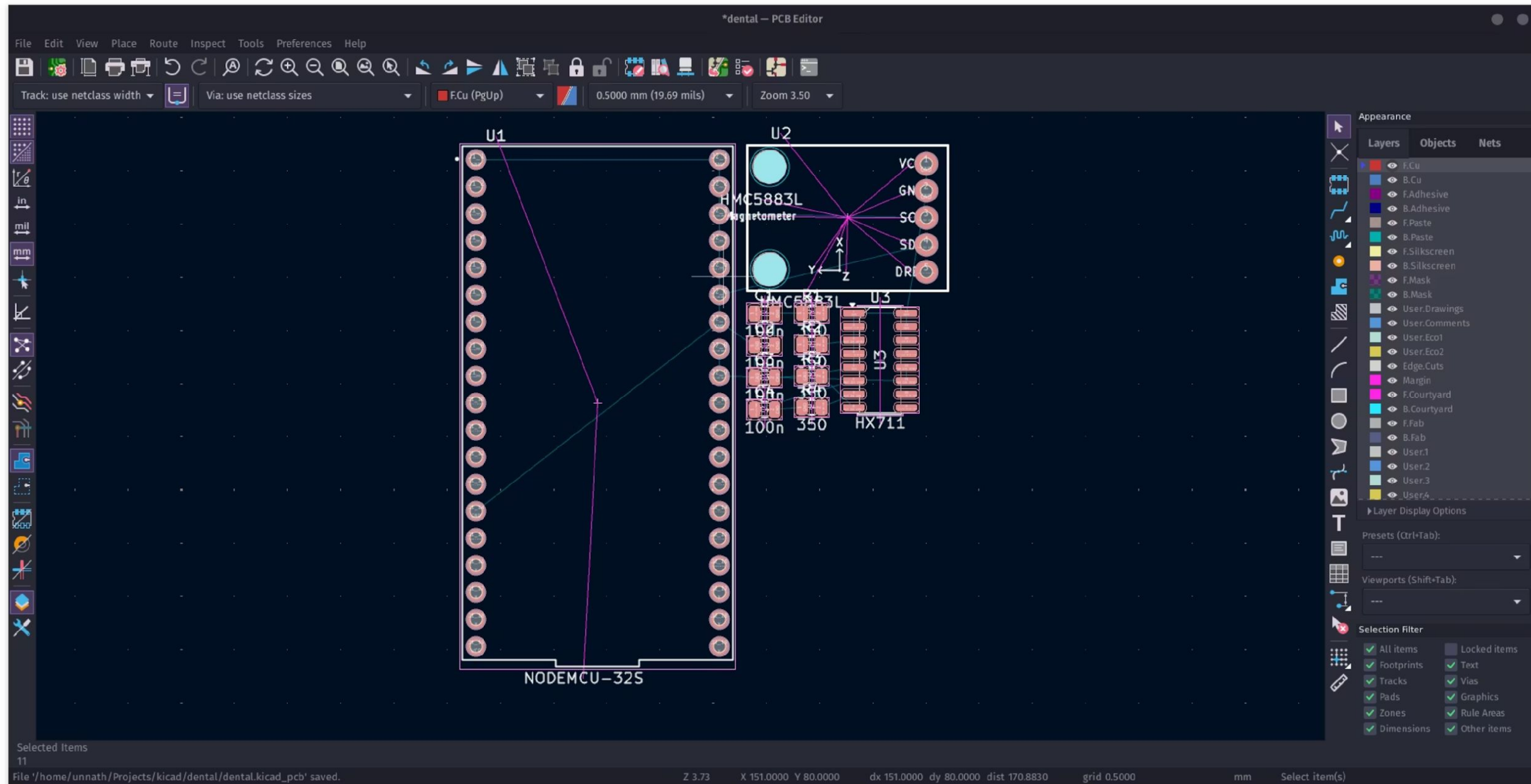
All electronics are enclosed within the Invisalign-like mould for maximum comfort and protection from saliva.



5cm x 5cm board, can be reduced further to 2~3cm x 2~3cm using only the MCU of ESP32 (this is just a prototype)



Schematic Design : KiCad (not final)



PCB Layout: KiCAD (not final)



Patient Comfort & Usability



Quick Calibration

Quick or NO magnetometer baseline establishment before sleep—no complex setup procedures required



Safe Materials

Medical-grade coated NdFeB magnet with no interference to braces, implants, or fillings



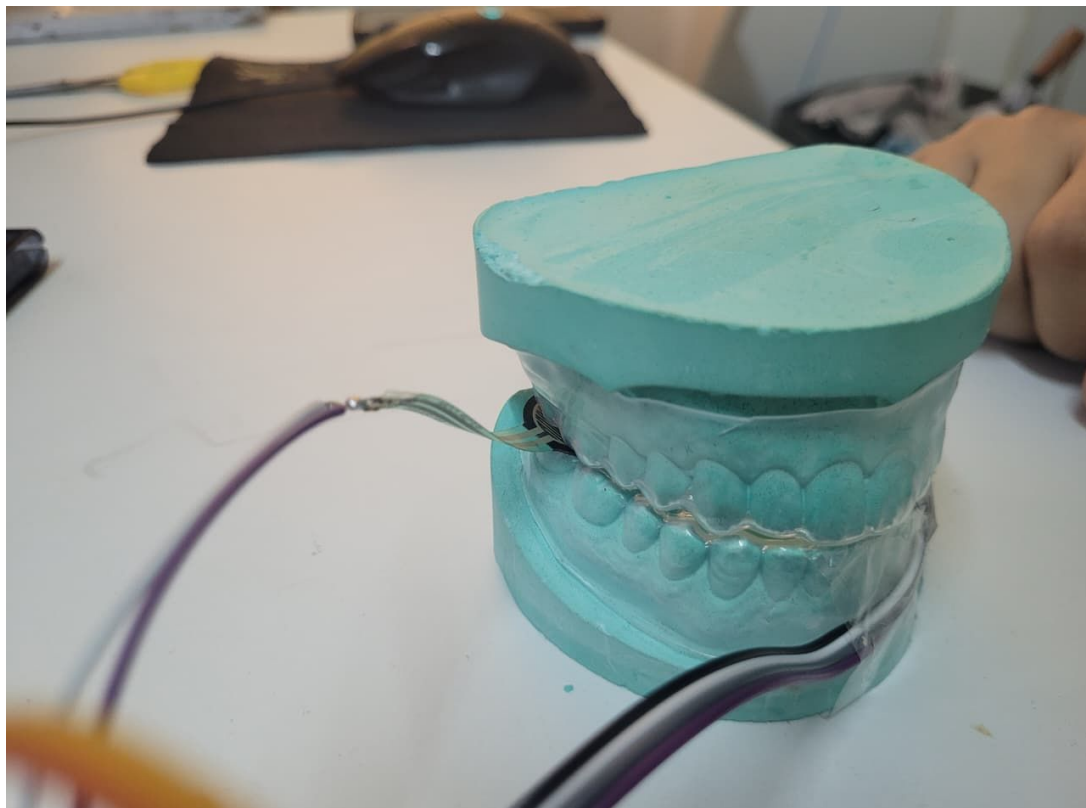
Fully Reusable

Durable appliance design for nightly use with simple cleaning protocols



Minimal Protrusion

Components integrated within appliance profile for natural feel and unobstructed speech



Data Pipeline & Clinical Metrics



Raw Signals

3-axis magnetometer data and optional FSR force readings captured at high frequency



Signal Processing

Windowed feature extraction including dominant frequency, band ratios, and amplitude measures



Classification

Machine learning models categorize activity into **rest, chewing, or grinding states (suggestions welcome)**



Clinical Output

Actionable metrics for dentists including **episode count, intensity, duration, and pressure heatmaps**

Machine Learning Performance

Models Evaluated

We trained and compared multiple classification approaches on labeled jaw motion data collected using a plastic denture model.

- Random Forest

Ensemble decision tree approach with strong baseline performance

- XGBoost

Gradient boosting with small model size and fast inference—ideal for deployment

- 1D CNN

Deep learning approach capturing temporal patterns in magnetometer signals

Key Results

Real-world mixed activity testing achieved **89-90% accuracy** with particularly strong discrimination between chewing and grinding events.

The model that performs best (1D-CNN on Real Data) selected for deployment due to its optimal balance of accuracy, model size, and computational efficiency on resource-constrained devices.

However, the **accuracy for RandomForest and XGBoost is also very similar** thus, it can be **deployed on ESP32 Hardware using TinyML!**

Based on power consumption requirements, we can choose to stream over BLE to a phone or run the model at sensor-level itself.

A **single 320 mAH LiPO battery** (small and convenient) lasts up to **5-6 hours** with the current setup (can be optimized further)

Visualization & FSR

Placement Plan

Real-Time Movement Reconstruction

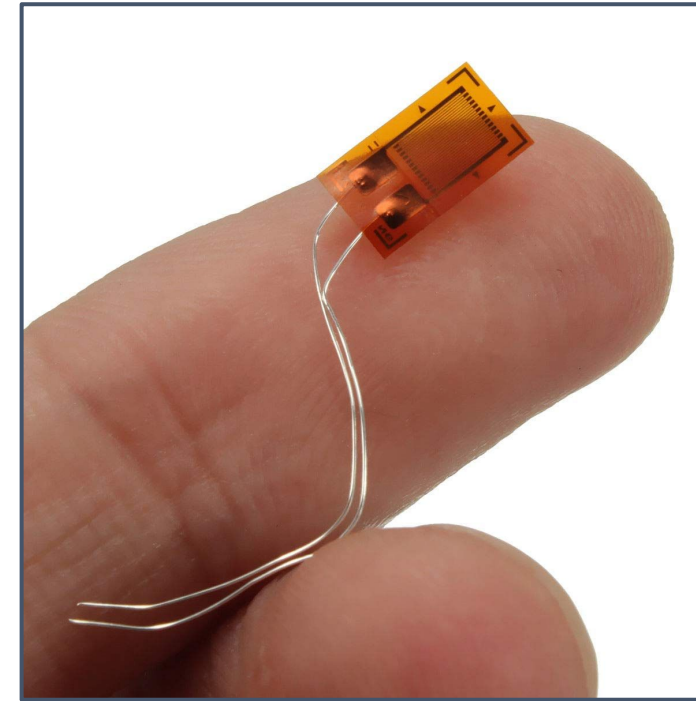
Our system generates 3D jaw movement animations from magnetometer data, enabling clinicians to visualize bruxism patterns. Blender-based rendering provides intuitive playback of grinding versus chewing versus rest states.

Current: Single-Tooth FSR

Prototype demonstrates ability to capture grinding intensity at one strategic location under the appliance. Provides single teeth heatmap and extra features for classification!

After This Meet: Multi-Zone Array

Custom PCB with integrated amplifiers will enable comprehensive **per-tooth pressure heatmapping**. Sensors will be chosen based on required level of precision



Micro-Strain Gauge:

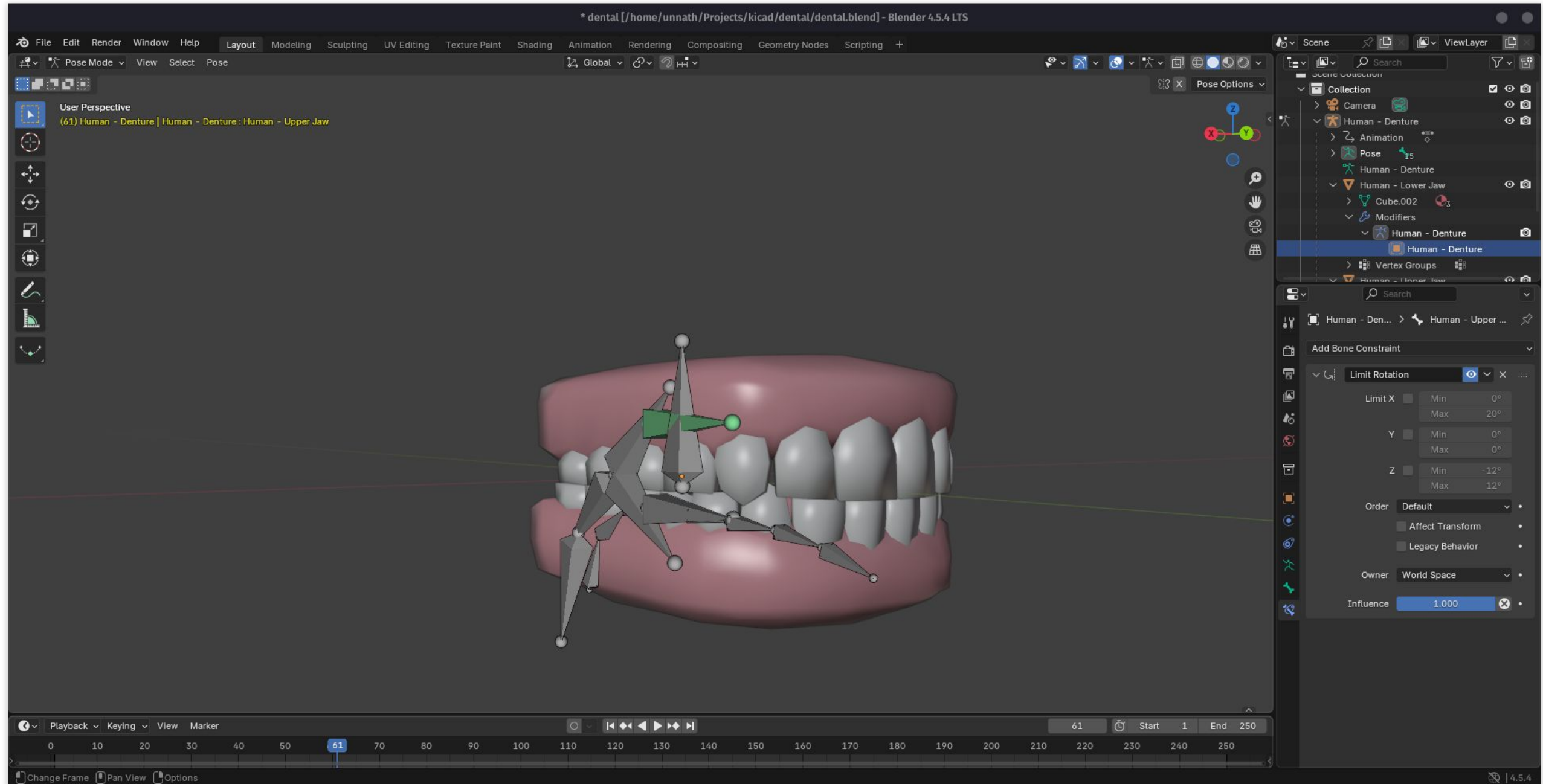
More resolution but high noise and requires amplifier.



Force Sensing Resistor:

Off the shelf, good for detection of clenching, moderate resolution

3D Blender Visualization



Clinical Impact & Next Steps

For Dentists

Objective quantification replaces guesswork. Identify which teeth sustain the most damage and optimize treatment planning for splints, aligners, and TMJ therapy.

For Patients

Comfortable home-based monitoring with long-term behavioral insights—no bulky headgear or expensive sleep lab visits required.

Development Roadmap

1

Current Phase

Magnetometer tracking, single FSR, XGBoost classification, visualization pipeline operational

2

Near-Term

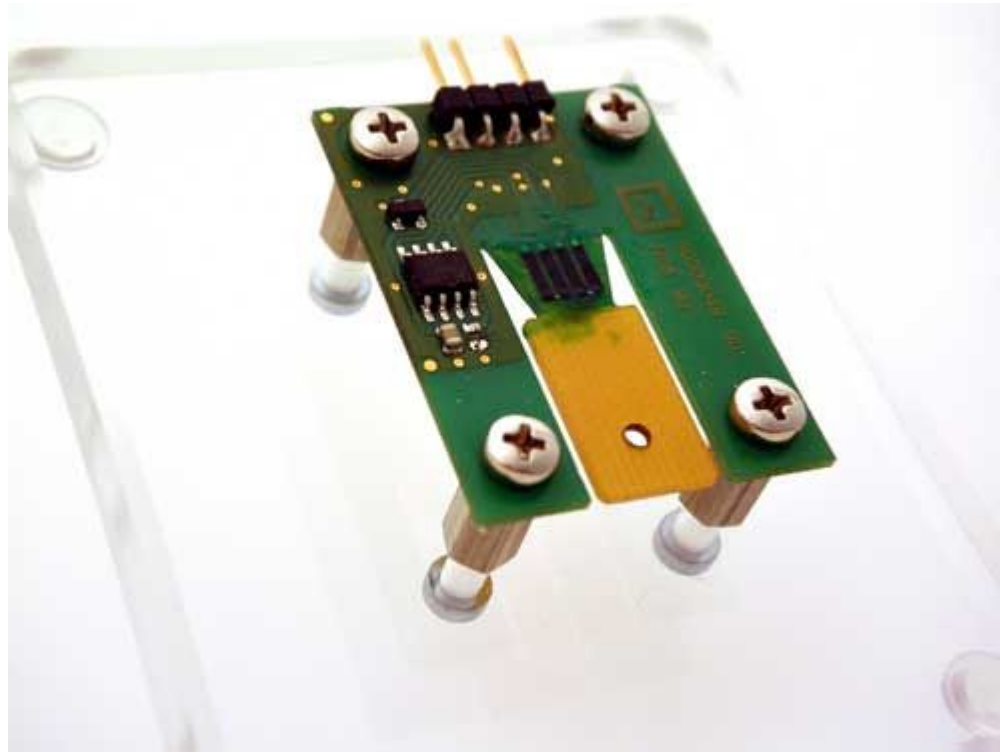
Multi-tooth FSR PCB, real-time on-device inference, mobile app development

3

Clinical Validation

Patient comfort studies, clinician dashboard, trials at partner dental hospital

"Our goal is to provide dentists with the first affordable, objective, and patient-friendly intra-oral bruxism monitoring system."



Similar Patents:

These are PCB/Plastic strain gauge solutions that can be fabricated per tooth and integrated with the mold.

Have to be custom produced so is not an “off-the-shelf” solution. But customization enables us to experiment more?

Your thoughts on this?